

Line Loss Costs/Benefits Calculation Documentation

1 Introduction

The `line_loss_costs.py` script calculates line loss costs for greenfield projects and line loss benefits for reconductoring projects. For greenfield projects, line losses represent a cost (energy that must be generated to compensate). For reconductoring projects, reduced losses compared to the old conductor represent a benefit.

2 Method Overview

2.1 Purpose

This script calculates line loss costs/benefits by:

1. Determining if project is greenfield or reconductoring
2. For greenfield: Calculating losses and converting to costs
3. For reconductoring: Comparing old vs. new conductor losses to calculate benefits
4. Computing present values using appropriate discount rates

2.2 Calculation Approach

Greenfield Projects:

$$\text{Annual Loss Cost} = \text{Losses (MWh/year)} \times \text{Electricity Price (\$/MWh)} \quad (1)$$

$$\text{Lifetime Nominal} = \text{Annual Cost} \times \text{Project Lifetime} \quad (2)$$

$$\text{Present Value} = \sum_{t=1}^N \frac{\text{Annual Cost}}{(1 + r)^{t+\text{start_year}-1}} \quad (3)$$

Reconductoring Projects:

$$\text{Loss Reduction} = \text{Old Losses} - \text{New Losses} \quad (4)$$

$$\text{Annual Benefit} = \text{Loss Reduction} \times \text{Electricity Price} \quad (5)$$

$$\text{Lifetime Benefit} = \text{Annual Benefit} \times \text{Project Lifetime} \quad (6)$$

$$\text{Present Value} = \sum_{t=1}^N \frac{\text{Annual Benefit}}{(1 + r)^{t+\text{start_year}-1}} \quad (7)$$

The script uses a normalized comparison method that accounts for capacity differences between old and new configurations.

2.3 Inputs and Outputs

Inputs:

- Project technical details (construction type, AC/DC, capacity, conductor type, reconductoring status)
- Old configuration details (for reconductoring: old capacity, old conductor type, old AC/DC)
- Physical details (line length)
- Circuit and resistance details
- Electricity price
- Social discount rate
- Project lifetime, delay years, construction years

Outputs:

- Annual cost/benefit
- Lifetime nominal cost/benefit
- Present value cost/benefit

2.4 Integration with Other Scripts

This script integrates with:

- `energy_losses.py` — `get_total_energy_losses` (primary config), `load_physical_details`, `load_circuit_and_resistance_details`, `calculate_phase_current`, `full_load_adjusted`, `calculate_line_losses`
- `smart_loaders.py` — `load_financing_social_discount_rate`, `load_project_technical_details`, `get_project_data_raw`
- `financial_utils.py` — `calculate_present_value`, `calculate_cod_year`
- `calculation_utils.py` — `to_percent`, `from_percent`, `build_category_string`, `calculate_conversion`
- `constants.py` — `HOURS_PER_YEAR`
- `smart_output.py` — `CTCCOutputManager` (`add_line_loss_costs`, `write_batch_summary`)

3 Key Functions

3.1 load_project_details()

Loads project technical details with line_loss_costs specific fields.

Returns:

- Multiple values including construction type, AC/DC, capacity, conductor type, utilization, reconductoring status, timeline parameters

3.2 calculate_configuration_losses(...)

Calculates line losses (conductor only) for a given configuration. Used for reconductoring baseline (old) and for greenfield comparison configurations. Optional `voltage_kv_override` allows using a different voltage (e.g., reconductoring: old tower voltage).

Parameters:

- `construction_type`, `ac_dc`, `capacity_mw`, `conductor_type`, `converter_type`, `line_utilization_percent`, `project_lifetime`
- `voltage_kv_override` (optional): If set, used instead of category lookup voltage

Returns:

- `losses_mwh_per_year`, `lifetime_losses_mwh` (conductor losses only; converter losses computed separately when needed)

Key Logic:

```
1 category = build_category_string(construction_type, ac_dc,
2   capacity_mw, conductor_type, converter_type)
3 line_length = load_physical_details()
4 circuit = load_circuit_and_resistance_details(category)
5 voltage_kv = voltage_kv_override if voltage_kv_override is not None
6   else circuit.voltage_kv
7 capacity_mw_str = f"{capacity_mw}MW"
8 phase_current = calculate_phase_current(capacity_mw_str, voltage_kv,
9   circuit.number_of_phases, circuit.number_of_circuits_poles,
10  ac_dc)
11 full_load_adj = full_load_adjusted(line_utilization_percent)
12 losses_mwh_per_year, lifetime_losses_mwh, ... =
13   calculate_line_losses(phase_current, full_load_adj, circuit.
14   AC_75_resistance, circuit.DC_20_resistance, ac_dc, ...)
```

3.3 main()

Main execution function that orchestrates the line loss cost/benefit calculation.

Workflow for Greenfield:

1. Gets losses via `get_total_energy_losses()` (line + converter)
2. Converts to costs ($\text{losses} \times \text{baseline_electricity_price}$), `start_year = calculate_cod_year(dela construction_years)`
3. Calculates present value; writes results (including optional line/converter breakdown) to CSV

Workflow for Reconductoring:

1. Baseline (old) losses: `calculate_configuration_losses(old...)` + `calculate_converter_losses` for old config; new (primary) losses: `get_total_energy_losses()`
2. Counterfactual: old conductor at new capacity via `calculate_configuration_losses` with new capacity
3. Benefit = (baseline or counterfactual total losses - new total losses) $\times$ price; PV and CSV as benefits

4 Example

The following example demonstrates a typical usage:

```

1 # Example 1: Greenfield Project
2 # 500 MW Overhead AC line
3 #
4 # Inputs:
5 #   - Capacity: 500 MW
6 #   - Line utilization: 70%
7 #   - Line length: 100 miles
8 #   - Losses: 5,000 MWh/year
9 #   - Electricity price: $50/MWh
10 #   - Project lifetime: 50 years
11 #   - Start year: 3 (after 2-year construction)
12 #
13 # Calculation:
14 #   Annual cost = 5,000 * $50 = $250,000/year
15 #   Lifetime nominal = $250,000 * 50 = $12.5M
16 #   PV = $250,000 * PV annuity (50 years, 3%, start year 3) = $8.2M
17 #
18 # Output:
19 #   - Annual cost: $250,000
20 #   - Lifetime nominal: $12.5M
21 #   - Present value: $8.2M
22 #
23 # Example 2: Reconductoring Project
24 # Upgrade 200 MW line to 350 MW
25 #

```

```

26 # Inputs:
27 #   - Old: 200 MW, Standard Aluminum, losses: 3,000 MWh/year
28 #   - New: 350 MW, Advanced Aluminum, losses: 4,000 MWh/year
29 #   - Counterfactual (old conductor @ 350 MW): 5,500 MWh/year
30 #   - Electricity price: $50/MWh
31 #
32 # Calculation (Normalized Method):
33 #   Loss reduction = 5,500 - 4,000 = 1,500 MWh/year
34 #   Annual benefit = 1,500 * $50 = $75,000/year
35 #   Lifetime benefit = $75,000 * 50 = $3.75M
36 #   PV = $75,000 * PV annuity (50 years, 3%) = $1.8M
37 #
38 # Output:
39 #   - Annual benefit: $75,000
40 #   - Lifetime benefit: $3.75M
41 #   - Present value: $1.8M

```

Integration Example:

```

1 # When called from ctcc.py (main() builds results):
2 csv_manager = CTCCOutputManager()
3 results = {
4     "annual_cost": total_annual_cost,
5     "total_nominal": total_lifetime_nominal,
6     "total_afudc": 0,
7     "total_pv": total_pv_loss_cost,
8     "line_loss_mwh_yr": losses_mwh_per_year,
9     "converter_loss_mwh_yr": total_converter_losses_mwh,
10    "total_loss_mwh_yr": total_losses_mwh_per_year,
11    "line_annual_cost": ..., "converter_annual_cost": ...,
12    "line_cost_pv": ..., "converter_cost_pv": ...,
13 }
14 csv_manager.add_line_loss_costs(results)
15 csv_manager.write_batch_summary()

```

5 Key Implementation Details

5.1 Greenfield vs. Reconductoring

The script handles two distinct cases:

- **Greenfield:** Line losses are a cost (energy that must be generated)
- **Reconductoring:** Reduced losses are a benefit (compared to old conductor)

5.2 Comparison Methods for Reconductoring

The script implements three comparison methods:

1. **Direct Comparison:** Old losses @ old capacity vs. new losses @ new capacity
2. **Counterfactual Comparison:** Old conductor @ new capacity vs. new conductor @ new capacity (used for output)
3. **Normalized Comparison:** Loss percentages weighted by delivered energy

The normalized method is used for the final output as it accounts for capacity differences.

5.3 Counterfactual Baseline

For reconductoring, the script calculates what losses would be if the old conductor were used at the new capacity. This provides a fair comparison by isolating the effect of the conductor upgrade.

5.4 Present Value Timing

Costs/benefits start when the project becomes operational (after delay and construction periods). The present value calculation discounts annual payments back to the base year.

5.5 AFUDC Treatment

Line loss costs/benefits are not AFUDC-eligible because they are operational costs/benefits, not construction costs.